

HW # 07 Solutions

Problem 1:

- (a) P_1 is positive semi-definite because

$$\lambda_{\min}(P_1) = 0$$
$$P_1 = N^T N, \text{ where } N = \begin{bmatrix} 1 & 3 \end{bmatrix}.$$

Also OK to say $N = \begin{bmatrix} 0 & 0 \\ 1 & 3 \end{bmatrix}$. In fact, there are an infinite number of possible answers. If you multiply N by an arbitrary orthogonal matrix, you can check that the matrix is still a square root.

Remark: Our results on the Schur complement provide Necessary and Sufficient Conditions for positive definite matrices. You can use them to prove that this matrix is NOT positive definite, but you cannot use them to prove that the matrix is semi-definite. Why? Because not positive definite also includes matrices that are indefinite, such as P_3 .

- (b) P_2 is positive definite because

$$\lambda_{\min}(P_2) = 0.1856$$
$$P_2 = N^T N, \text{ where } N = \begin{bmatrix} -0.3767 & 0.1409 & 0.1543 \\ -0.0255 & -0.7991 & 0.6671 \\ 3.2401 & 3.6886 & 4.5417 \end{bmatrix}.$$

Other answers are possible, such as

$$N = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

Remark: It is perfectly fine to show that P_2 is positive definite using the results on Schur complements, as in Problem 2.

- (c) P_3 is not positive semi-definite because

$$\lambda_{\min}(P_3) = -2.9165 < 0$$

Problem 2:

- (a) $P_1 > 0 \Leftrightarrow 1 > 0$ and $8 - 3(1)3 > 0$. But $8 - 9 = -1 < 0$

$\therefore$ Not Positive Definite.

- (b) $P_2 > 0 \Leftrightarrow A = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} > 0$ and $10 - \begin{bmatrix} 6 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ 7 \end{bmatrix} > 0$.

$$A > 0 \text{ is obvious, } 10 - \begin{bmatrix} 6 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} = -38.25$$

$\therefore$ Not Positive Definite.

$$(c) P_3 > 0 \Leftrightarrow \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} > 0 \text{ and } a - \begin{bmatrix} 6 & 7 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ 7 \end{bmatrix} > 0.$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} > 0 \text{ because } 1 > 0 \text{ and } 5 - 4 > 0$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}^{-1} = \frac{1}{1} \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$$

$$a - \begin{bmatrix} 6 & 7 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} > 0 \text{ if and only if } a - \begin{bmatrix} 6 & 7 \end{bmatrix} \begin{bmatrix} 16 \\ -5 \end{bmatrix} = a - 61 > 0$$

$$\therefore \boxed{a > 61}$$

Problem 3: For $\langle x, y \rangle = x^\top Q y$, the minimum norm solution is

$$\hat{x} = Q^{-1} A^\top (A Q^{-1} A^\top)^{-1} b$$

(a) $Q = I$, and we have

$$\hat{x} = \begin{bmatrix} -0.0952 \\ 0.0476 \\ 0.4762 \end{bmatrix}$$

(b) $Q \neq I$, and we have

$$\hat{x} = \begin{bmatrix} -0.6497 \\ 0.3248 \\ 0.3376 \end{bmatrix}$$

Problem 4:

(a)

$$\hat{x} = \begin{bmatrix} 0.6195 \\ 0.4591 \end{bmatrix}, \text{ Covariance } \Sigma = \begin{bmatrix} 4.0000 & -2.7500 \\ -2.7500 & 2.0000 \end{bmatrix},$$

$$\hat{K} = \begin{bmatrix} -2.0000 & 1.0000 \\ 1.5000 & -0.5000 \end{bmatrix}$$

(b)

$$\hat{x} = \begin{bmatrix} -1.4303 \\ 1.8791 \end{bmatrix}, \text{ Covariance } \Sigma = \begin{bmatrix} 0.0679 & -0.0260 \\ -0.0260 & 0.1129 \end{bmatrix},$$

$$\hat{K} = \begin{bmatrix} -0.1050 & -0.0525 & 0.1895 \\ 0.1872 & 0.1564 & -0.1313 \end{bmatrix}$$

(c)

$$\hat{x} = \begin{bmatrix} -1.2201 \\ 1.5368 \end{bmatrix}, \text{ Covariance } \Sigma = \begin{bmatrix} 0.0487 & 0.0054 \\ 0.0054 & 0.0618 \end{bmatrix}$$

$$\hat{K} = \begin{bmatrix} -0.0540 & 0.1046 & 0.1480 & -0.0518 \\ 0.1041 & 0.0715 & -0.0637 & 0.0843 \end{bmatrix}$$

Problem 5:

(a) From the handout, we identify

$$\begin{aligned}
 X_1 &= \begin{bmatrix} X \\ Y \end{bmatrix}, \quad X_2 = Z, \\
 \Sigma_{11} &= \begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix}, \quad \Sigma_{12} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \Sigma_{21} = \Sigma_{12}^\top, \quad \Sigma_{22} = 2 \\
 \mu_1 &= \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad \mu_2 = [1] \\
 \therefore \mu_{1|2} &= \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (z - \mu_2) \\
 &= \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} (z - 1) \\
 &= \begin{bmatrix} -\frac{3}{2} + \frac{z}{2} \\ z - 1 \end{bmatrix} \\
 \Sigma_{1|2} &= \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} = \begin{bmatrix} \frac{3}{2} & 1 \\ 1 & 2 \end{bmatrix}
 \end{aligned}$$

(b) We have

$$\begin{aligned}
 \mu &= \mu_{1|2} = \begin{bmatrix} -\frac{3}{2} + \frac{z}{2} \\ z - 1 \end{bmatrix} \\
 \Sigma &= \Sigma_{1|2} = \begin{bmatrix} \frac{3}{2} & 1 \\ 1 & 2 \end{bmatrix}
 \end{aligned}$$

We apply the handout to compute

$$\begin{aligned}
 \mu_{1|2} &= \left(-\frac{3}{2} + \frac{z}{2} \right) + (1) \left(\frac{1}{2} \right) (y - (z - 1)) = -1 + \frac{y}{2} \\
 \Sigma_{1|2} &= \frac{3}{2} - (1) \left(\frac{1}{2} \right) (1) = \frac{3}{2} - \frac{1}{2} = 1
 \end{aligned}$$

$\therefore$ The random variable $X|_{Z=z}$ conditioned on $Y|_{Z=z} = y$ has a normal distribution with

$$\begin{aligned}
 \text{mean} &= -1 + \frac{y}{2} \\
 \sigma^2 &= 1
 \end{aligned}$$

(c) We go back to the beginning and identify

$$\begin{aligned}
 X_1 &= X, \quad X_2 = \begin{bmatrix} Y \\ Z \end{bmatrix}, \quad x_2 = \begin{bmatrix} y \\ z \end{bmatrix} \\
 \Sigma_{11} &= 2, \quad \Sigma_{12} = \begin{bmatrix} 2 & 1 \end{bmatrix}, \quad \Sigma_{21} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \Sigma_{22} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
\mu_1 &= -1 \text{ and } \mu_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
\therefore \mu_{1|2} &= \mu_1 + \Sigma_{12}\Sigma_{12}^{-1} \left(\begin{bmatrix} y \\ z \end{bmatrix} - \mu_2 \right) \\
&= -1 + \begin{bmatrix} \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} y \\ z-1 \end{bmatrix} \\
&= -1 + \frac{y}{2} \\
\sigma^2 &= 2 - \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 1
\end{aligned}$$

(d) They are the same !

Problem 6: We give two approaches to part (a)

- (a) **Method I:** We apply Gram-Schmidt to build an orthonormal basis for M_k , that is $M_k = \text{span}\{v^1, \dots, v^k\}$ with $\langle v^i, v^j \rangle = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$.

We then know that

$$\hat{x}_k = \hat{\alpha}_1 v^1 + \dots + \hat{\alpha}_k v^k$$

where $\hat{\alpha}_i = \langle x, v^i \rangle$.

Because $y_{k+1} \perp M_k = \text{span}\{v^1, \dots, v^k\}$, we define

$$v^{k+1} = \frac{y_{k+1}}{\|y_{k+1}\|}$$

Hence, we have

$$\hat{x}_{k+1} = \hat{\alpha}_1 v^1 + \dots + \hat{\alpha}_k v^k + \hat{\alpha}_{k+1} v^{k+1}$$

where $\hat{\alpha}_i = \langle x, v^i \rangle$.

Therefore,

$$\begin{aligned}
\hat{x}_{k+1} &= \hat{x}_k + \hat{\alpha}_{k+1} v^{k+1} \\
&= \hat{x}_k + \hat{\alpha}_{k+1} \frac{y_{k+1}}{\|y_{k+1}\|} \\
\therefore \hat{x}_{k+1} &= \hat{x}_k + \beta y_{k+1}
\end{aligned}$$

where

$$\begin{aligned}
\beta &= \frac{1}{\|y_{k+1}\|} \langle x, v^{k+1} \rangle \\
&= \frac{1}{\|y_{k+1}\|^2} \langle x, y_{k+1} \rangle \\
&= \frac{\langle x, y_{k+1} \rangle}{\langle y_{k+1}, y_{k+1} \rangle}.
\end{aligned}$$

Method II: The solution to $\hat{x}_k = \operatorname{argmin}_{m \in M_k} \|x - m\|$ is

$$\hat{x}_k = \alpha_{k,1}y_1 + \alpha_{k,2}y_2 + \dots + \alpha_{k,k}y_k$$

where the parameter vector $\alpha_k = [\alpha_{k,1} \ \alpha_{k,2} \ \dots \ \alpha_{k,k}]^\top$ satisfies the normal equation

$$G_k^\top \alpha_k = \beta_k$$

and $[G_k]_{i,j} = \langle y_i, y_j \rangle$, $[\beta_k]_i = \langle x, y_i \rangle$, for $i, j = 1, 2, \dots, k$.

Similarly, the solution to $\hat{x}_{k+1} = \operatorname{argmin}_{m \in M_{k+1}} \|x - m\|$ is

$$\hat{x}_{k+1} = \alpha_{k+1,1}y_1 + \alpha_{k+1,2}y_2 + \dots + \alpha_{k+1,k}y_k + \alpha_{k+1,k+1}y_{k+1}$$

where $\alpha_{k+1} = [\alpha_{k+1,1} \ \alpha_{k+1,2} \ \dots \ \alpha_{k+1,k} \ \alpha_{k+1,k+1}]^\top$ satisfies the normal equation

$$G_{k+1}^\top \alpha_{k+1} = \beta_{k+1}$$

and $[G_{k+1}]_{i,j} = \langle y_i, y_j \rangle$, $[\beta_{k+1}]_i = \langle x, y_i \rangle$, for $i, j = 1, 2, \dots, k+1$.

We note that $y_{k+1} \perp M_k \Rightarrow \langle y_{k+1}, y_i \rangle = 0$ for $i = 1, 2, \dots, k$, and thus the normal equation partitions as $G_{k+1}^\top \alpha_{k+1} = \beta_{k+1}$ into

$$\left[\begin{array}{ccc|c} \langle y_1, y_1 \rangle & \cdots & \langle y_1, y_k \rangle & 0 \\ \vdots & \ddots & \vdots & \vdots \\ \langle y_k, y_1 \rangle & \cdots & \langle y_k, y_k \rangle & 0 \\ \hline 0 & \cdots & 0 & \langle y_{k+1}, y_{k+1} \rangle \end{array} \right] \left[\begin{array}{c} \alpha_{k+1,1} \\ \vdots \\ \alpha_{k+1,k} \\ \alpha_{k+1,k+1} \end{array} \right] = \left[\begin{array}{c} \langle x, y_1 \rangle \\ \vdots \\ \langle x, y_k \rangle \\ \langle x, y_{k+1} \rangle \end{array} \right],$$

which is equivalent to

$$\left\{ \begin{array}{l} G_k^\top [\alpha_{k+1}]_{1:k} = \beta_k \\ \langle y_{k+1}, y_{k+1} \rangle \alpha_{k+1,k+1} = \langle x, y_{k+1} \rangle \end{array} \right.$$

Thus, $[\alpha_{k+1}]_{1:k} = (G_k^\top)^{-1} \beta_k = \alpha_k$ and $\alpha_{k+1,k+1} = \frac{\langle x, y_{k+1} \rangle}{\langle y_{k+1}, y_{k+1} \rangle}$, leading to

$$\begin{aligned} \hat{x}_{k+1} &= \alpha_{k,1}y_1 + \alpha_{k,2}y_2 + \dots + \alpha_{k,k}y_k + \frac{\langle x, y_{k+1} \rangle}{\langle y_{k+1}, y_{k+1} \rangle} y_{k+1} \\ &= \hat{x}_k + \frac{\langle x, y_{k+1} \rangle}{\|y_{k+1}\|^2} y_{k+1}. \end{aligned}$$

(b) From the Projection Theorem,

$$y_{k+1} - \hat{y}_{k+1|k} \perp M_k$$

moreover, $\hat{y}_{k+1|k} \in M_k$, and thus

$$\begin{aligned} M_{k+1} &= \operatorname{span}\{y_1, \dots, y_k, y_{k+1}\} \\ &= \operatorname{span}\{y_1, \dots, y_k, y_{k+1} - \hat{y}_{k+1|k}\} \\ &= \operatorname{span}\{y_1, \dots, y_k, \tilde{y}_{k+1}\} \end{aligned}$$

where $\tilde{y}_{k+1} = y_{k+1} - \hat{y}_{k+1|k}$ is orthogonal to M_k . Hence, from (a),

$$\hat{x}_{k+1} = \hat{x}_k + \beta \tilde{y}_{k+1}$$

where $\beta = \frac{\langle x, \tilde{y}_{k+1} \rangle}{\langle \tilde{y}_{k+1}, \tilde{y}_{k+1} \rangle}$.

Substituting in for $\tilde{y}_{k+1}$, we have

$$\hat{x}_{k+1} = \hat{x}_k + \beta(y_{k+1} - \hat{y}_{k+1|k})$$

where

$$\beta = \frac{\langle x, y_{k+1} - \hat{y}_{k+1|k} \rangle}{\langle y_{k+1} - \hat{y}_{k+1|k}, y_{k+1} - \hat{y}_{k+1|k} \rangle} = \frac{\langle x, y_{k+1} - \hat{y}_{k+1|k} \rangle}{\|y_{k+1} - \hat{y}_{k+1|k}\|^2}$$

□